Dynamic Pages

Session 12



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Objectives

- *Explain the working of dynamic pages*
- *Explain the working of CGI*
- *Describe the configuration process of Apache for supporting CGI*
- *Explain Server Side Includes*
- *Describe the configuration process of Apache for Server Side Includes*
- *Describe the configuration process of Apache for Directory Indexing*

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Common Gateway Interface (CGI)

- Protocol that defines the communication standard between the Web server and an external program
- A CGI program (CGI script)
  - Can be written in many languages, such as Perl, Python, PHP, or C
  - Stored in a directory called `cgi-bin` present on the Web server
- Used to:
  - Create interactive Web documents
  - Insert data into databases
  - Use a given database of information to generate pages
  - Resolve form data
  - Respond to a form entry with e-mail or another Web page
  - Set and read cookies
  - Calculate hit counts
  - Monitor Web traffic

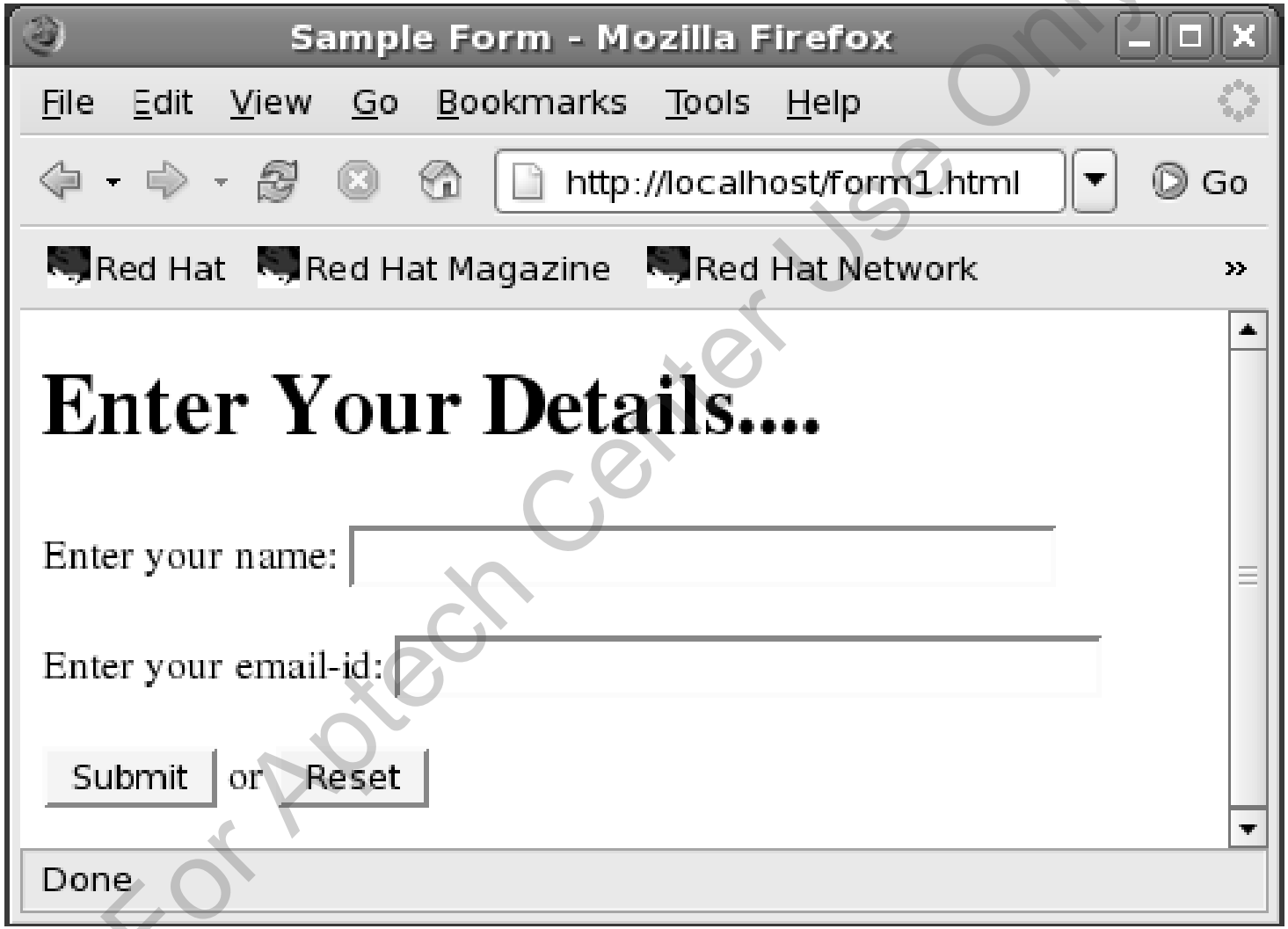
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Snippet

```
<html>
  <head>
    <title>Sample Form</title>
  </head>
  <body><h1>Enter Your Details....</h1>
    <form action=”perl_script.cgi” method=post>
      Enter your name:
      <input type=”text” name=”Name” size=30 maxlength=50><p>
      Enter your email-id:
      <input type=”text” name =”Email” size=30 maxlength=50> <p>
      <input type =”submit” value = “Submit”>
      or
      <input type=”reset” value=”Reset”> <p>
    </form>
  </body>
</html>
```

Snippet displays an example where a sample form accepts user data in the HTML file

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HTML Form

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- Sample form accepts user data in the HTML file named `form1.html`
- On clicking the submit button, a Perl script will be used to process and display the data entered by the user
- The CGI scripts must be saved in the `/usr/local/apache2/cgi-bin` directory to be executed on the Web browser
- In addition, the `httpd.conf` file must contain the following code to enable Apache server to execute CGI scripts from the `cgi-bin` directory

```
ScriptAlias /cgi-bin/ /usr/local/apache2/cgi-bin/
```

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Directives Used to Configure Apache for CGI

- `ScriptAlias` directive
- `Options` directive
- `SetHandler` directive
- `AddHandler` directive

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ScriptAlias Directive

- Defines a directory for executing CGI programs
- `mod_alias` module implements the directive
- Can be used for directories that are outside the `documentroot`

Syntax

```
ScriptAlias URL-path file path | directory path
```

Snippet

```
ScriptAlias /cgi-prog/ /usr/local/apache2/cgi-prog/
```

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Options Directive

- Features applicable to a directory can be specified
- Can be used with the `Directory` directive to configure Apache server for CGI execution

Syntax

```
Options [+|-]option [[+|-]option] ...
```

Snippet

```
<Directory /usr/local/apache2/htdocs/mydir>
Options +ExecCGI
</Directory>
```

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Options Directive

Options	Functions
ExecCGI	Enables the execution of CGI scripts
FollowSymLinks	Configures the Apache Web server to follow symbolic links. A symbolic link is a special file in Linux that contains the pathname to another file
Includes	Enables Server-side includes
IncludesNOEXEC	Enables Server-side includes but disables the #exec cmd and #exec cgi commands
Indexes	Returns a formatted listing of the directory when a directory does not contain a DirectoryIndex directive in that directory (The DirectoryIndex directive defines a list of files that must be searched when a client requests a directory)
MultiViews	Enables content-negotiated multiviews. Content negotiation enables Apache server to select the representation of a resource based on the media type, languages, character set, and encoding information provided by the browser
SymLinksIfOwnerMatch	Enables Apache server to follow symbolic links only for the target file where the user ID of the file is the same as the link
None	Disables all options
All	Enables all the options except for MultiViews

Different Options and their functions

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SetHandler Directive

- Defines a handler for a particular directory
- Must be specified along with the `ExecCGI` inside a directory to enable CGI execution
- Overrides default handlers; therefore, it ignores the activities, such as, interpreting URLs ending in a slash(/) as directories

Syntax

```
SetHandler handler_name|None
```

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AddHandler Directive

- Enables the administrator to specify the filename extensions to the specified handler
- Must use it along with `Options` directive inside a directory section

Syntax

```
AddHandler handler_name extension
```

Snippet

```
<Directory "/usr/local/apache2/cgidir">
  Options +ExecCGI
  SetHandler cgi-script
  AddHandler cgi-script .cgi .pl .py
</Directory>
```

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Server Side Includes (SSI)

- Directives placed in HTML pages, which are executed on the server while the pages are being served
- Server-side scripting language that adds small pieces of dynamic content, such as:
  - Current date and time
  - Document name
  - Content length
- Have an extension of `.shtml`

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Server Side Includes (SSI)

- Used to:
  - Provide interactivity to a Website
  - Help in maintaining the Website
  - Add a common piece of code
  - Control directives in SSI that can include conditional navigation menus
- Features:
  - Supports only one data type: text
  - Has a simple design
  - Is easy to learn
  - Is used on Web servers, such as Apache, lighttpd, and IIS

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Directives Used to Configure Apache for SSI



- `AddType` Directive
- `AddOutputFilter` Directive
- `AddInputFilter` Directive

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AddType Directive

- Links a filename extension to the given content type

Syntax

```
AddType MIME_type file_extension
```

Snippet

```
AddType image/gif .gif
```

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AddOutputFilter Directive

- Links the specified filename extension to the specified filter
- A filter processes the data transmitted between the server and client
- When a client transfers data to a server, the input filter processes this data
- On the other hand, when the server sends data back to the client, the output filter processes it

Syntax

```
AddOutputFilter filter file_extension
```

Snippet

```
AddOutputFilter INCLUDES .shtml
```

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AddInputFilter Directive

- Directs the filename extension to the filters that will respond to the client requests and `POST` input when Apache processes them

Syntax

```
AddInputFilter filter file_extension
```

- The `file_extension` argument is case-insensitive and can be specified with or without a leading dot
- Filenames may have multiple extensions and the `file_extension` argument will be compared against each of them

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SSI Commands

- Generate dynamic information in HTML pages
- When the server retrieves an SSI document, it executes the SSI commands, and performs the specified action

Syntax

```
<!-- #command attribute=value attribute=value … -->
```

- Some of the SSI commands are
  - `echo` Command
  - `Include` File
  - `Include` Virtual
  - `printenv`
  - `set` Command

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echo Command

- Adds contents of an HTTP environment variable or the include variables, such as:
  - DATE_GMT
  - LAST_MODIFIED
  - DOCUMENT_NAME
- For example, to display the document name, enter the code in an HTML file

Snippet

```
The name of this file is <!--#echo var="DOCUMENT_NAME" -->
```

Option	Description
CONTENT_LENGTH	Specifies the total number of data bytes sent by the POST method to the server
DOCUMENT_ROOT	Specifies the root directory from where the site pages are served
DATE_LOCAL	Specifies the current date and time
LAST_MODIFIED	Specifies the date and time when the document was last modified
HTTP_USER_AGENT	Specifies the browser name

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Include File

- Used to include the contents of a file that is placed in the same directory as the file containing the `include` command, or a subdirectory of it
- It will not allow an absolute path
- For example, to include a file containing the contact address, enter:

Snippet

```
<!--#include file="contactadrr.txt" -->
```

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Include Virtual

- Includes a file whose path is relative to the document root
- For example, to include a `file`, `test.txt`, enter:

Snippet

```
<!--#include virtual="/usr/local/apache2/includes/test.txt" -->
```

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printenv

- Lists all the existing variables and their corresponding values
- It includes both environment and user-defined variables
- This command does not have any attributes

Syntax

```
printenv
```

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set Command

- Defines the value of a variable and has two attributes:
  - Var
  - Value
- Attributes of Set command

Attributes	Description
var	Defines the name of the variable
value	Specifies the value to be set for the variable

Snippet

```
<!--#set var="category" value="help" -->
```

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Directory Indexing

- Displays the list of included directories instead of the actual Web pages
- Two different ways:
  - Generate a file containing the list of all the directories by using the `DirectoryIndex` directive
  - Configure Apache Web server to generate a list of all the directories using the `mod_autoindex` module
- Directory indexing directives:
  - `DirectoryIndex` Directive
  - `Automatic Index Generation`

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DirectoryIndex Directive

- Defines a list of files that must be searched when a client requests a directory

Syntax

```
DirectoryIndex filename1 filename2 ...
```

Snippet

```
DirectoryIndex index.html
```

- If a request, `http://yourserver/dirname/`, is send, then the server will redirect to `http://myserver/dirname/index.html`

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Automatic Index Generation

- The Options directive can be used to configure Apache to generate indexes
- To configure Apache Web server to generate indexes, enter the code:

Snippet

```
Options +Indexes
```

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Summary

- Dynamic Web page interacts with users and displays dynamic content
- CGI is a specification that transmits information between a CGI program and the Web
- A CGI program is a script written in scripting languages that accepts and returns data, and conforms to the CGI specification
- SSI is used as an alternative to CGI programming for implementing dynamic content into HTML pages
- Directory indexing displays the list of included directories instead of the actual Web pages

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